

# **Computer Architectures**

## **Exam of 18.2.2026 - part I**

### **Question #1B**

Let consider the *Hazards* that may arise in a pipelined processor. You are requested to

1. List the different types of Hazards, detailing in particular the different categories of Data Hazards
2. Describe for every type and category an example where the Hazard happens
3. Identify the corresponding data dependency (for Data Hazards, only).

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### Question #2B

Let consider the Branch Prediction Mechanism based on the Branch History Table (BHT).

You are requested to

1. Describe the architecture of a BHT
2. Describe the behavior of a BHT, detailing when the processor accesses it, how, and which information it gets from it

Assuming that the processor uses 32 bit addresses, each instruction is 4 byte wide, the BHT is composed of 8 1-bit entries, and is initially empty (i.e., full of 0s, meaning NotTaken), consider the following instructions executed in sequence (for each instruction the corresponding address is reported)

```

0x00B80250  lw   x1, 0(x2)
0x00B80254  bne  x1, x10, lab1  (branch taken, jumping to 0x00008000)
0x00008000  addi x2, x2, #1
0x00008004  beq  x2, x11, lab2   (branch not taken)
0x00008008  andi x5, x5, #1
0x0000800C  beq  x5, x12, lab3   (branch taken, jumping to 0x0000AA10)
0x0000AA10  add  x1, x2, x3
0x0000AA14  beq  x1, x13, lab4   (branch taken, jumping to 0x000040C8)
0x000040C8  lw   x6, 4(x2)
0x000040CC  addi x2, x2, #1
0x000040D0  bne  x2, x10, lab5   (branch taken, jumping to 0x00009200)
0x00009200  add  x4, x5, x0
0x00009204  beq  x4, x11, lab6   (branch taken, jumping to 0x0000AAF0)
0x0000AAF0  sw   x4, 8(x2)
0x0000AAF4  beq  x4, x12, lab1   (branch taken, jumping to 0x00008000)
  
```

Using the following tables, you are asked to

3. Determine which of the branch instructions were correctly predicted, and which were mispredicted
4. Identify the final content of the BHT.

| Address    | Instruction       | Addressed BHT line | Prediction (T/NT) | Result (T/NT) | Prediction (correct/wrong) |
|------------|-------------------|--------------------|-------------------|---------------|----------------------------|
| 0x00B80250 | lw x1, 0(x2)      |                    |                   |               |                            |
| 0x00B80254 | bne x1, x10, lab1 |                    |                   |               |                            |
| 0x00008000 | addi x2, x2, #1   |                    |                   |               |                            |
| 0x00008004 | beq x2, x11, lab2 |                    |                   |               |                            |
| 0x00008008 | andi x5, x5, #1   |                    |                   |               |                            |
| 0x0000800C | beq x5, x12, lab3 |                    |                   |               |                            |
| 0x0000AA10 | add x1, x2, x3    |                    |                   |               |                            |
| 0x0000AA14 | beq x1, x13, lab4 |                    |                   |               |                            |
| 0x000040C8 | lw x6, 4(x2)      |                    |                   |               |                            |
| 0x000040CC | addi x2, x2, #1   |                    |                   |               |                            |
| 0x000040D0 | bne x2, x10, lab5 |                    |                   |               |                            |
| 0x00009200 | add x4, x5, x0    |                    |                   |               |                            |
| 0x00009204 | beq x4, x11, lab6 |                    |                   |               |                            |
| 0x0000AAF0 | sw x4, 8(x2)      |                    |                   |               |                            |
| 0x0000AAF4 | beq x4, x12, lab1 |                    |                   |               |                            |

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| Initial BHT content |         |
|---------------------|---------|
| Index               | Content |
| 0                   | 0       |
| 1                   | 0       |
| 2                   | 0       |
| 3                   | 0       |
| 4                   | 0       |
| 5                   | 0       |
| 6                   | 0       |
| 7                   | 0       |

| Final BHT content |         |
|-------------------|---------|
| Index             | Content |
| 0                 |         |
| 1                 |         |
| 2                 |         |
| 3                 |         |
| 4                 |         |
| 5                 |         |
| 6                 |         |
| 7                 |         |